
Shreya Bharathwajan

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EDUCATION

University of California - San Diego (UCSD)

Bachelor of Science in Neurobiology and Data Science

Sept 2023 - June 2026 (expected)

La Jolla, CA

WORK EXPERIENCE

Undergraduate Researcher at the Department of Pediatrics

UC San Diego Health – Briganti Lab

Sept 2025 - Present

La Jolla, CA

- Perform **RNA sequencing (RNA-seq)**, Cell culture, Western blots, PCR and **bioinformatics analyses** to study **post-transcriptional gene regulation**, focusing on **microRNA signalling**, **transcriptomic variation**, and **alternative splicing** in cardiomyopathy.
- Study **miR-214-mediated regulation** and **PLN mutation mechanisms** in dilated cardiomyopathy using **iPSCs** and **stem cell-derived cardiomyocytes**.
- Develop and implement **data preprocessing, statistical modelling, and visualisation pipelines** using **Python (Pandas, NumPy, SciPy, Matplotlib)** and **R**, improving analytical reproducibility and data integrity

Data Analyst and Experimental Research Assistant, Systems Neuroscience VR Lab

UCSD Cognitive Science Department

Dec 2024 - Present

La Jolla, CA

- Lead **spatial cognition experiments** as Head of the Experimental Team; design and analyse **VR-based studies** using **eye-tracking, motion sensors, and behavioral data**.
- Apply **machine learning and computer vision models** (e.g., **Fréchet Inception Distance, trajectory similarity metrics**) to quantify **spatial navigation performance**.
- Develop and maintain **data pipelines** for large multimodal datasets; build **3D environments** in **Blender, Unreal Engine, and Fusion 360**.
- Recipient of **Nick Spitzer's award for Innovation in Neuroscience Research**, with a grant of \$5000 for Individual project

Research Intern

ThinkNeuro, LLC

Jun 2025- Sept 2025

CA

- Led a **national research team** investigating **EEG-based motor imagery classification** using **CNNs** and **deep learning**, achieving **~80% accuracy** across sessions.
- Analysed **multisession EEG datasets** with **Python** and **R** to study **neural plasticity** and **BCI model generalisation**.
- Presented findings at the **UC Berkeley Virtual Research Symposium 2025**, recognised as a **Top 20 project nationwide** in **computational neuroscience**.

Residential Assistant

UC San Diego Residence Life

Aug 2023 - Present

La Jolla, CA

- Facilitate **community building, conflict resolution, and mentorship** across 70+ residents, promoting inclusive living environments and leadership development.

TECHNICAL SKILLS AND ANALYTICAL SKILLS

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- **Programming:** Python, R, Java, SQL, Git, Jupyter, VS Code
 - **Data Science & ML:** TensorFlow, Keras, Scikit-learn, Hugging Face, OpenCV, NLP, CNN, LSTM, KNN, PCA
 - **Bioinformatics:** RNA-Seq, DESeq2, ggplot2, NCBI BLAST, UniProt, UCSC Genome Browser, AlphaFold2, Pathway Analysis, Functional Genomics, Phylogenetics, PyMOL, Docker, PLINK
 - **Lab Techniques:** CRISPR-Cas9, PCR, qPCR, RT-PCR, Western Blot, Cell Culture, Electrophoresis, Microscopy, Spectrophotometry, Chromatography, iPSC, DNA/RNA extraction, Cloning
 - **Software & Visualisation:** Matplotlib, Seaborn, Fusion 360, Blender, Unreal Engine, Tableau, Power BI, Seaborn

PROJECTS

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- **EEG + EMG Wheelchair (Triton Neurotech):** Developing a hybrid brain-muscle interface system enabling movement control via **EMG** and communication via **EEG**.
 - **RNA-Seq Alzheimer's Biomarker Project:** Performed **DESeq2-based differential expression, PCA, and heatmap visualisation** to identify potential genetic biomarkers.
 - **KNN Image Classifier:** Implemented an **OOP-based KNN algorithm** in **Python** for image recognition and distance-based classification.
 - **DeepLabCut Behavioural Analysis:** Used **DeepLabCut** and **OpenCV** for **motion tracking and kinematic feature extraction** from behavioural datasets of mice.
 - **Song Recommender System:** Built a **data-driven recommendation model** in **Python (Pandas, NumPy)** using **cosine similarity** and **visual analytics**.